NON-ACUTE AND ACUTE PORPHYRIAS

Patients with all forms of non-acute porphyria manifest cutaneous findings predominantly on sun-exposed body areas. The non-acute porphyrias include **porphyria cutanea tarda (PCT)**, the most prevalent type of porphyria overall; **erythropoietic protoporphyria (EPP)**; **congenital erythropoietic porphyria (CEP)**; and **hepatoerythropoietic porphyria (HEP)**, the homozygous variant of PCT. These disorders are characterized by variable cutaneous features, including mild to severe photosensitivity, increased skin fragility, vesicles and bullae, burning and stinging sensations, edema, pruritus, hypertrichosis, scarring, hyperpigmentation, and mild to severe scleroderma-like changes.

The acute porphyrias include **acute intermittent porphyria (AIP)**, **variegate porphyria (VP)**, **hereditary coproporphyria (HCP)**, and **ALA dehydratase deficiency porphyria (plumboporphyria)**. AIP is the most common acute porphyria worldwide, although VP predominates in certain regions such as Chile and South Africa. Acute porphyrias are primarily associated with neurovisceral symptoms, including episodic abdominal pain, which can be mistaken for surgical emergencies such as acute appendicitis or diverticulitis. Treatment with analgesics may prolong or worsen the attack in some cases.

In addition to neurovisceral manifestations, patients with VP and HCP may also develop cutaneous symptoms—such as photosensitivity, skin fragility, bullous lesions on sun-exposed areas, erosions, scarring, and pigmentary changes—due to accumulation of photosensitizing porphyrins. Therefore, VP and HCP are classified as **neurocutaneous porphyrias**, distinguishing them from AIP and ALA dehydratase deficiency porphyria, which do not involve cutaneous abnormalities.

Acute attacks can be precipitated by various factors, including certain drugs (e.g., barbiturates) and general anesthesia. A particularly serious complication is **muscle weakness**, which may range from mild involvement of small muscle groups to flaccid paralysis affecting multiple limbs, leading to paraplegia and respiratory compromise. This can be accompanied by severe sensory loss and amyotrophy. In some cases, paralysis progresses rapidly and may involve cranial nerves or resemble Guillain-Barré syndrome.

Ethanol and estrogens are known triggers of acute attacks in AIP, VP, and HCP. These agents can also exacerbate PCT, although acute neurovisceral attacks do not occur in PCT, and the underlying mechanisms are likely different.

Pathophysiology of Cutaneous Signs and Symptoms in the Non-Acute Porphyrias

Photosensitivity is the hallmark cutaneous manifestation in non-acute porphyrias. It occurs in PCT, EPP, CEP, HEP, and also in VP and HCP—disorders in which both cutaneous and neurovisceral symptoms may be present. The essential factor underlying photosensitivity is elevated levels of **plasma and tissue porphyrins**, which act as potent photosensitizers due to their unique photobiologic and spectroscopic properties.

In contrast, patients with AIP or ALA dehydratase deficiency porphyria accumulate **aminolevulinic acid (ALA)** and **porphobilinogen (PBG)**—non-photosensitizing precursors—and therefore do not develop photosensitivity.

Patients with erythropoietic porphyrias, particularly EPP, often report intense burning pain and pruritus during or after sunlight exposure. A "priming" effect has been described in EPP, where an initial subthreshold sun exposure appears to enhance the skin's response to subsequent exposure the following day, suggesting impaired or delayed repair of porphyrin-mediated photodamage. These painful symptoms are typically absent in chronic hepatic porphyrias like PCT and VP.

While the mechanisms of photosensitivity and sclerodermoid changes in hepatic porphyrias are partially understood, the causes of **hyperpigmentation**, **hypopigmentation**, and **hypertrichosis** remain unclear.